Ground-state measure transfer: the slice identity is exact, the large-field estimate remains open

Later corrections: the operator-inequality note separates measure from operator control; the corrected Jacobian note and instability note withdraw the Hamiltonian truncation and sharp-crossover claims; the lattice note treats a different, single-step regime (relevant passages read).

> **Correction (2026-10-02).** The original title and lead claimed that isotropic Wilson-measure estimates transfer verbatim to the Kogut–Susskind ground state and that a normalized large-field bound with running coupling is thereby available. Those claims are **withdrawn**. The fixed-temporal-spacing slice identity is proved below; its Wilson-to-Hamiltonian limit is conditional on ground-state projector convergence, with no interchange of limits asserted. Section 3 retains a continuum action lower bound, not a probability bound. The asserted universal constant $c = 8\pi^2$, the sharp crossover $\eta_* = 2g^2/c$, and the claim that the good region is almost everything at that crossover are withdrawn. The stray $\cdot 2/1$ in the action display is replaced by its explicit coefficient $2/g^2$. Section 4 also incorporates the later corrections to the flow normalization and to the distinction between kernel tails and Hamiltonian errors.

On a **fixed finite spatial lattice**, the square of the normalized Perron vector of a positive transfer matrix is its infinite-time Euclidean slice marginal. For the Kogut–Susskind Hamiltonian the analogous identity holds directly for its heat semigroup. Identifying these two marginals as the temporal spacing tends to zero requires a separate convergence result. Neither identity transfers an estimate proved only for an isotropic Wilson measure without control of that limit. Neither identity proves a large-field estimate.

The available calculation here is instead the following: with the continuum normalization $S_E/\hbar = (4g^2)^{-1}\int |F|^2$, a four-dimensional block of side ℓ with root-mean-square field at least η/ℓ^2 forces action at least $\eta^2/(4g^2)$. This scale cancellation does not control a normalized partition-function ratio. The proposed suppression $C\exp[-c\eta^2/g(\ell)^2]$ remains a **heuristic target** for the interacting theory. No local large-field closure, Hamiltonian decimation bound, or continuum mass gap is proved.

1. The transfer matrix and the two limits

Fix $G = SU(N)$, $g > 0$, spatial spacing $a > 0$, and a finite periodic spatial box. Let $d\mu$ be normalized product Haar measure on its spatial links. Write a_t for the temporal spacing **in length units** (the physical time step is a_t/c), and $\epsilon = a_t/a$. In the convention of the obligations map, §1 (passage read), the anisotropic Wilson action is

$$\frac{S_E}{\hbar} = \frac{2}{g^2}\left[\epsilon^{-1}\sum_{P\ \text{temporal}}\left(N - \operatorname{Re}\operatorname{tr} U_P\right) + \epsilon\sum_{P\ \text{spatial}}\left(N - \operatorname{Re}\operatorname{tr} U_P\right)\right].$$

The target Hamiltonian, on the gauge-invariant subspace, is

$$H_{\mathrm{KS}} = \frac{\hbar c}{a}\left[\frac{g^2}{2}\sum_e(-\Delta_e) + \frac{2}{g^2}\sum_p\left(N - \operatorname{Re}\operatorname{tr} U_p\right)\right].$$

The transfer-matrix references are Lüscher, *Commun. Math. Phys.* **54** (1977) 283–292, and Creutz, *Phys. Rev. D* **15** (1977) 1128–1136 (both metadata level; not used here as a checked proof of temporal-limit uniformity). A character-expansion motivation of this Hamiltonian limit is not a ground-state convergence theorem. The former formula $\mathcal{T} = \exp[-a_t H_{\mathrm{KS}}/\hbar](1 + O(a_t))$

is withdrawn: it omitted c with the length convention, specified no operator topology, and supplied no spectral convergence control.

Proposition 1 (fixed temporal spacing). Fix $a_t > 0$. Assume the physical transfer matrix $\mathcal{T}_{a_t}$ is compact, self-adjoint, nonnegative, and has a simple largest eigenvalue $\lambda_{a_t} > 0$ with normalized positive eigenvector Ω_{a_t}. These are the transfer-matrix hypotheses used below; they are distinct from T1 for H_{KS}. For bounded multiplication F and any boundary vector χ with $\langle \Omega_{a_t}, \chi\rangle \neq 0$,

$$\langle \Omega_{a_t}, F\Omega_{a_t}\rangle = \lim_{n\to\infty} \frac{\langle \mathcal{T}_{a_t}^n \chi, F\mathcal{T}_{a_t}^n \chi\rangle}{\|\mathcal{T}_{a_t}^n \chi\|^2}.$$

For nonnegative χ and the Wilson transfer-kernel representation, the ratio is the expectation in the anisotropic Euclidean slab of $2n$ steps, with boundary weights χ and F inserted at the middle slice. Thus its limit is the marginal $|\Omega_{a_t}|^2 d\mu$. For a general signed or complex χ the spectral formula still holds, but its boundary functional need not be a positive Euclidean probability measure.

Proof. Let P_{a_t} project onto Ω_{a_t}. Compactness and simplicity give

$$r_{a_t} = \|\mathcal{T}_{a_t}(1 - P_{a_t})\|/\lambda_{a_t} < 1, \qquad \lambda_{a_t}^{-n}\mathcal{T}_{a_t}^n \chi = P_{a_t}\chi + O(r_{a_t}^n \|\chi\|)$$

in Hilbert norm. Since $P_{a_t}\chi \neq 0$, the normalized ratio converges to the displayed expectation. Integrating successive transfer kernels is the slab functional by its construction. □

Hamiltonian identity and conditional Wilson limit. The finite-lattice H_{KS} has a unique positive ground state Ω and a gap by T1 of the obligations map, §2 (passage read). Spectral projection therefore gives, directly,

$$\langle \Omega, F\Omega\rangle = \lim_{\tau\to\infty} \frac{\langle e^{-\tau H_{\mathrm{KS}}/\hbar}\chi, Fe^{-\tau H_{\mathrm{KS}}/\hbar}\chi\rangle}{\|e^{-\tau H_{\mathrm{KS}}/\hbar}\chi\|^2},$$

where τ is physical time and $\langle \Omega, \chi\rangle \neq 0$. This is an exact semigroup identity, without invoking a Wilson limit.

To identify it with the preceding anisotropic Wilson marginal, a sufficient additional hypothesis is

$$\|P_{a_t} - P_0\| \longrightarrow 0 \quad (a_t \downarrow 0), \qquad P_0 = |\Omega\rangle\langle\Omega|.$$

Under this hypothesis the positive normalized vectors converge in $L^2(d\mu)$, and their densities converge in $L^1(d\mu)$, since

$$\big\| |\Omega_{a_t}|^2 - |\Omega|^2 \big\|_1 \leq 2\|\Omega_{a_t} - \Omega\|_2.$$

Consequently the **iterated** limit, first $n \to \infty$ at fixed a_t, then $a_t \downarrow 0$, equals $\langle \Omega, F\Omega\rangle$ for every fixed bounded F. Projector convergence is not proved here. No interchange with the reverse order, no joint limit, and no estimate uniform in spatial volume or spacing is asserted.

2. What transfers, and what does not

At fixed a_t, a probability bound for a property of a single slice under the infinite-time **anisotropic** Wilson measure is exactly a bound under $|\Omega_{a_t}|^2 d\mu$. Under the additional projector convergence hypothesis, any bound

$$\int_A |\Omega_{a_t}|^2 d\mu \leq p \qquad (0 < a_t < a_{t,0})$$

for a fixed measurable slice event A, with the same p, passes to $\int_A |\Omega|^2 d\mu \leq p$. One may also start with slab bounds uniform in n and a_t and then take the indicated iterated limits. If the event or the coupling varies in the limit, its convergence must be controlled separately.

An isotropic estimate has $a_t = a$, whereas the Hamiltonian limit takes $a_t/a \to 0$ at fixed a. It is not a bound for that family. Nor is a four-dimensional block event automatically a property

of its middle slice: temporal coherence and a spatial-slice field constraint are different events. A comparison of ensembles and observables is required.

Plaquette factorization of the unnormalized Wilson weight does not make plaquettes independent: they share links, and normalization involves all links. It does not supply local probability control. The identities note, §§4–5 (passages read), shows why one particular eigenvalue-equation comparison loses local control through an extensive energy threshold. The present identity provides another representation; it **does not close** that missing local estimate or prove that Hamiltonian methods cannot obtain one.

3. Action cost at a block scale, not a probability theorem

Proved continuum action bound. For a smooth field with finite Euclidean action in the stated normalization, let B be a four-dimensional cube of volume ℓ^4 and define the gauge-invariant root-mean-square field

$$\mathcal{F}_B = \left(\ell^{-4}\int_B |F|^2 d^4x\right)^{1/2}.$$

On $\mathcal{F}_B \geq \eta/\ell^2$, positivity gives

$$\frac{S_E}{\hbar} = \frac{1}{4g^2}\int |F|^2 d^4x \geq \frac{1}{4g^2}\int_B |F|^2 d^4x \geq \frac{\eta^2}{4g^2}.$$

The norm sums the colour and ordered spacetime indices in the action. The same argument applies to a flowed block L^2 constraint if the flow exists and decreases this action. It is the lower-bound argument of the action note, §2 (passage read); its further sharpness and probability claims are not used. There is no ℓ in this bound. There is also no lattice-spacing parameter in this continuum calculation: an exact Wilson-lattice bound requires an explicit lattice field definition and its action comparison. No supported constant-field sharpness or uniform lattice result is inferred from the continuum display.

Missing normalization. For any finite regulated probability measure $d\nu$ and dimensionless action s, the interacting law is $d\mathbb{P} = Z^{-1}e^{-s}d\nu$. An action bound $s \geq q$ on A gives only

$$\mathbb{P}(A) = \frac{\int_A e^{-s}d\nu}{Z} \leq \frac{e^{-q}\nu(A)}{Z}.$$

A useful local estimate needs a lower bound on the denominator relative to the constrained numerator, with controlled boundary, entropy, volume and cutoff dependence. The action bound alone supplies none. Accordingly

$$\mathbb{P}\left(\mathcal{F}_B \geq \frac{\eta}{\ell^2}\right) \leq C\exp\left[-\frac{c\eta^2}{g(\ell)^2}\right]$$

is a **heuristic target, not an available theorem in this note**. Replacing bare g by running $g(\ell)$ requires a renormalized measure and observable comparison as well; it is not licensed by scale cancellation. No citation checked here establishes this interacting bound with the needed constants and limit uniformity.

Free-field comparison, with its own observable. Gaussian tails do have quadratic exponents, but their constants depend on the covariance, smearing and threshold. For a scalar Gaussian X of variance σ^2, $\mathbb{P}(|X| \geq u) \leq 2e^{-u^2/(2\sigma^2)}$ by the exponential-moment bound. For the three-component, single-colour abelian magnetic field with Gaussian heat smearing used in the free-field calculation, §§5a–5c (passages read),

$$\widetilde{G}_{ij}(k) = \frac{g^2|k|}{2}e^{-2t|k|^2}(\delta_{ij} - \hat{k}_i\hat{k}_j).$$

Taking its trace and radial integral gives

$$\langle |b_t|^2 \rangle = \frac{g^2}{2\pi^2} \int_0^\infty k^3 e^{-2tk^2} dk = \frac{g^2}{16\pi^2 t^2}.$$

Rotational invariance makes each component's variance $\sigma^2 = g^2/(48\pi^2 t^2)$, one third of this trace. Their joint Gaussian law is $\sigma^2 I_3$. Its radial density is proportional to $r^2 e^{-r^2/(2\sigma^2)}$, so in the large-threshold regime

$$\log \mathbb{P}\left(|b_t| > \frac{\eta_t}{t} \right) = -\frac{24\pi^2 \eta_t^2}{g^2} + O(\log(\eta_t/g)), \qquad \eta_t/g \to \infty.$$

The polynomial radial prefactor is not a constant uniform in threshold. The former $8\pi^2$ comparison from the typical-field note, §3 (passage read), treated the trace variance as a scalar variance for the vector norm and cannot fix a constant here. Moreover, with $\ell^2 = 8t$, a threshold η_ℓ/ℓ^2 means $\eta_t = \eta_\ell/8$; even this particular free-field logarithmic exponent becomes $-3\pi^2\eta_\ell^2/(8g^2)$. Neither value is a sharp-block or an interacting constant. A sharp average uses a different covariance; nonabelian field averaging also requires a gauge convention or an invariant observable such as $\mathcal{F}_B$.

At the cutoff scale compact Wilson plaquettes have bounded action. That fact gives no useful normalized block-coherence estimate, and $\ell \gg a$ alone does not prove one either.

4. Where this leaves the route

Correction to the former competition. Neither Proposition 1 nor Section 3 controls the decimation error outside a proved rare set. The assertion that a crossover $\eta_* = 2g^2/c$ makes the good region almost everything is withdrawn. Even assuming the old bound with unit prefactor, its value there was $e^{-4g^2/c} \to 1$: an upper bound tending to one establishes no rarity. It also does not prove that the actual bad-set probability tends to one.

The corrected Jacobian note, §§2–3 (full-read), gives an **absolute kernel-tail bound for a specified auxiliary covariant linear evolution**, under smoothness and bounded curvature hypotheses. With its full curvature block norm $\Gamma(s)$, its budget is

$$A_t = 2\int_0^t \Gamma(s) ds, \qquad \delta(R,t) \le e^{A_t} \left[\operatorname{erfc}(\sqrt{2}\kappa) + \frac{2\sqrt{2}\kappa}{\sqrt{\pi}} e^{-2\kappa^2} \right], \qquad R = \kappa\sqrt{8t}.$$

If $\Gamma(s) \le \eta/\ell^2$ throughout $0 \le s \le t$ and $\ell^2 = 8t$, then $A_t \le \eta/4$, not 2η. A lower bound on a block RMS field does not give this curvature supremum upper bound. Even a proved probability bound for $\mathcal{F}_B$ therefore would not supply this kernel budget without an additional comparison.

Under the further hypothetical bound $\mathbb{P}(A_\eta) \le Ce^{-c\eta^2/g^2}$ and a controlled corresponding growth factor $e^{\eta/4}$, the product has schematic exponent $-c\eta^2/g^2 + \eta/4$, whose algebraic zero is $g^2/(4c)$. This is bookkeeping of additional assumptions, **not a sharp crossover or a probability-weighted Hamiltonian error theorem**. A tail bound for $\mathcal{F}_B$ alone does not justify that product, particularly when curvature is unbounded on the event.

For finite A_t, increasing the range κ controls the displayed kernel majorant. This is the surviving range observation behind the operator-inequality note, §1 (passage read), which supersedes the fixed-range competition. It does not prove a relative form bound for a conjugated Hamiltonian. Measure rarity also does not alone control a perturbation of unrestricted sign: that note's min-max and localized trial-state arguments, §3 (full-read), distinguish positive errors from negative wells. The later instability correction, §§2 and 4 (passages read), supplies no sharp lower bound on truncation tails. The separate single-lattice-step discussion in the lattice note, §§1–2 (passages read), must not be read as a fixed-physical-scale continuum estimate or as a repair of these missing operator comparisons.

5. Consequence for STATE

The usable result is the fixed-temporal-spacing spectral slice identity and its exact Hamiltonian-semigroup counterpart. Passing from the anisotropic Wilson family to the Kogut–Susskind marginal remains conditional here on projector convergence. Transferring a local bound then requires that bound for the same slice observable, uniformly in that family and in every further limit claimed.

The scale-invariant continuum action lower bound survives. The normalized interacting large-field estimate with running coupling, the import of isotropic estimates verbatim, and the claimed Hamiltonian decimation closure are withdrawn; the probability estimate is retained only as a heuristic target. The next required estimates are the normalized anisotropic slice bound with controlled limits and, for a Hamiltonian comparison, a signed operator or relative form bound. No advance on the continuum existence or physical mass gap follows from the marginal identity alone.